

Supporting Information for

Indirect reciprocity under opinion synchronization

Yohsuke Murase and Christian Hilbe

Corresponding author: Yohsuke Murase
E-mail: yohsuke.murase@gmail.com

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1. Models

A. Models of indirect reciprocity. We consider a population of size $N(\gg 1)$ in which each player is engaged in the donation game repeatedly. In each round, a player is randomly selected as a donor, and another one is selected randomly as a recipient. The donor chooses either to cooperate (C) or defect (D). By cooperating, the donor pays a cost c to benefit the recipient by $b(>c)$. By defecting, players get no benefit or pay no cost. The game is repeated for sufficiently many rounds changing the donor and recipient in each round, and the players accumulate their payoffs over the rounds.

During this sequence of the donation games, players form opinions about each other. The opinion player i holds about j is denoted as m_{ij}, and the entire matrix M is referred to as an image matrix. Following the convention of previous literature, opinions are binary, either good (G) or bad (B), and they can change over time. Donor players condition their actions on the opinions they hold about the recipient. The way they decide whether to cooperate in the donation game is determined by their action rule P. The action is observed by other players, who then update their opinions about the donor with their assessment rules R. The combination of these rules $\{P, R\}$ is referred to as the strategy or the social norm.

We consider the so-called stochastic 'second-order' norms. Donor i decides its action $A_{ij}(\in \{C, D\})$ based on their opinions about the recipient j. The donor cooperates with the probability prescribed by the action rule $P(m_{ij})$. The assessment by an observer k on the donor i is determined by the action taken by the donor and the observer's opinion about the recipient. The assessment rule R is a function of m_{kj} and A_{ij} that returns the probability that the donor is assessed as G. The new opinion about the donor m'_{ki} is G with probability $R(m_{kj}, A_{ij})$ otherwise $m'_{ki} = B$. Since neither P nor R depends on the self-image, m_{ii} does not play any role in this model.

Both the players' actions and their assessments can be subject to errors. First, we consider implementation errors. Such errors affect donors who wish to cooperate; with probability μ_e such donors defect by mistake. As a result, instead of their intended strategy $P(X)$, players implement the effective strategy

$$\tilde{P}(X) = (1 - \mu_e)\, P(X). \qquad [1]$$

In this paper, we consider $\mu_e = 0$ unless explicitly stated because the dynamics are not qualitatively different from the case of $\mu_e = 0^+$. Similarly, we consider assessment errors. These errors happen when new reputations are assigned. With probability μ_a, the assignment is the opposite of those prescribed by the social norm. As a result, the effective assessment rules become

$$\tilde{R}(X, A) = (1 - \mu_a)\, R(X, A) + \mu_a \left[1 - R(X, A)\right]. \qquad [2]$$

In the following, we denote $\tilde{R}$ as R and $\tilde{P}$ as P for simplicity unless explicitly stated. In this paper, we limit ourselves to the cases that $\mu_a > 0$, i.e., $0 < R(X, A) < 1$ for any X and A. This condition is crucial to make the dynamics ergodic. Irrespective of the initial conditions, the system reaches a unique stationary state after a long enough time.

We consider four models of indirect reciprocity: the solitary observation model, the public assessment model, the simultaneous observation model, and the gossiping model. They differ in how the opinions are updated and how the assessments are made. The common part of the models, except for the gossiping model, is described by the pseudo-code in Algorithm 1. In each time step, a donor i and a recipient j are randomly chosen. The donor decides its action C or D based on its action rule $P_i(m_{ij})$. Then, the opinions are updated by a function UpdateOpinions(i, j), which is model-dependent, as shown in the following subsections.

Algorithm 1 A pseudo-code of a time step for the indirect reciprocity models. The function UpdateOpinions(i, j) updates the opinions based on the action taken by i toward j. Implementation of the function UpdateOpinions(i, j) is model-dependent.

```
1:  procedure Main
2:      for t_init times do
3:          ConductGame                                   ▷ Initial thermalization
4:      for t_max times do
5:          ConductGame                                   ▷ Main simulation
6:      Calculate the cooperation probabilities between players
7:  procedure ConductGame
8:      i ← Randomly select a donor
9:      j ← Randomly select a recipient other than i
10:     coopProb ← P_i(m[i][j])        ▷ Get the cooperation probability using i's action rule
11:     r ← Random(0, 1)
12:     if r < coopProb then
13:         action ← C                                    ▷ Cooperation
14:     else
15:         action ← D                                    ▷ Defection
16:     UpdateOpinions(i, j, action)
```

Algorithm 2 A pseudo-code of UpdateOpinions(i, j) for the solitary observation model.

```
procedure UpdateOpinions(i, j, action)
    k ← Randomly select an observer other than i, j
    goodProb ← R_k(m[k][j], action)          ▷ Assess using k's assessment rule
    r ← Random(0, 1)
    if r < goodProb then
        m[k][i] ← G
    else
        m[k][i] ← B
```

Algorithm 3 A pseudo-code of UpdateOpinions(i, j) for the public assessment model. Note that the opinions are always fully synchronized by model assumption. Namely, each column of the image matrix is always identical. All players use the same assessment rule.

```
procedure UpdateOpinions(i, j, action)
    goodProb ← R(m[i][j], action)
    r ← Random(0, 1)
    if r < goodProb then
        for k ← 0 to N − 1 do
            m[k][i] ← G
    else
        for k ← 0 to N − 1 do
            m[k][i] ← B
```

B. Solitary observation model. The first model we consider is the solitary observation model (1). In this model, a randomly selected observer k updates the opinion about the donor i based on the action taken by the donor i toward the recipient j. The observer k assesses the donor i using its assessment rule R_k and its opinion about the recipient m_{kj}. The pseudo-code is shown in Algorithm 2.

Alternatively, we may consider an equivalent model. Each time step consists of an interaction phase and an opinion update phase. In the interaction phase, for every pair of players i and j, the donor i conducts its action based on its action rule $P_i(m_{ij})$. After the interaction phase, the opinions are updated based on the actions taken by the donors. A player k updates the opinion about i, m_{ki}, based on a randomly selected action that i took. Let's say the action was conducted toward j. The opinion is updated based on the assessment rule $R_k(m_{kj}, A_{ij})$. In an opinion update phase, all elements of the image matrix are updated.

These two interpretations are essentially equivalent. The point is that $O(N^2)$ games are conducted to update $O(N^2)$ elements of the entire image matrix. Each action is observed only once by a single observer on average. This way, elements in each column are statistically independent, and thus analytical calculations are straightforward.

C. Public assessment model. In the public assessment model, everyone fully agrees on their opinions. Namely, each column of the image matrix is always identical. This is the opposite extreme of the solitary observation model. Since the opinions are always synchronized, the population has a single assessment rule R. A pseudo-code of the opinion update is shown in Algorithm 3.

This assumption simplifies the analysis of the reputation dynamics. In the literature, this model has been most intensively studied after a seminal work by Ohtsuki and Iwasa (2) on the leading eight. In one of those studies, Murase and Hilbe (3) generalized the model of Ohtsuki and Iwasa to incorporate stochastic action and assessment rules. For the generalized model, they identified the cooperative ESSs comprehensively by deriving the necessary and sufficient conditions for the social norms to be evolutionarily stable. In this paper, we will use this stochastic version of the model to compare the results with the other models.

D. Simultaneous observation model. In the simultaneous observation model, a single action is observed by other players, and the opinions about the donor are updated simultaneously (4). Unlike the solitary observation model, the number of observers is not limited to one. All players except the donor themselves observe the action A_{ij} with observation probability q. Each observer makes the assessments independently. Let us say k is the observer. This observer k updates the opinion about the donor m_{ki} based on the action A_{ij} and the observer's opinion about the recipient m_{kj}. With probability $R_k(m_{kj}, A_{ij})$, m_{ki} is updated to G. Otherwise, it is updated to B. Since the opinions toward the recipient may differ between observers and each assessment is subject to errors, the new opinions are not necessarily shared between the observers, even if they use the same assessment rule. This model includes the solitary observation model as a special case when $q = 1/N$. A pseudo-code of the opinion update is shown in Algorithm 4.

While this model is more realistic because multiple observers can assess the same action, it is more challenging to analyze than the solitary observation model since the opinions are not statistically independent. In general, it is hard to obtain

Algorithm 4 A pseudo-code of UpdateOpinions(i, j) for the simultaneous observation model. All players update their opinions about i simultaneously using their own assessment rule with observation probability q.

```
1: procedure UpdateOpinions(i, j, action)
2:     for k ← 0 to N − 1 do
3:         r ← Random(0, 1)
4:         if r < q then
5:             goodProb ← R_k(m[k][j], action)
6:             r2 ← Random(0, 1)
7:             if r2 < goodProb then
8:                 m[k][i] ← G
9:             else
10:                m[k][i] ← B
```

analytical solutions for the dynamics of the opinions except for a few special cases (5, 6). Simulations are required to investigate the dynamics of the opinions and the cooperation probabilities.

E. Gossiping model. Lastly, we consider a model of gossiping proposed by Kawakatsu et al. (7). This model uses an alternative version of the solitary observation model as a base. Namely, at each time step, donation games are played by all pairs of players. After the interaction phase, the opinions are updated. Each assessment is conducted according to an action randomly selected from the donor's actions. In addition to these interaction and the assessment phases, the gossiping phase is introduced, where players exchange their opinions with randomly selected peers. In a gossipnig event, three players i, j, and k are randomly selected, and m_{ij} is copied to m_{kj}. This event is repeated $\tau N^3/2$ times in a gossiping phase, where τ is the gossiping duration. A pseudo-code is shown in Algorithm 5. A source code is provided by the authors of the original paper (7).

This model falls back to the solitary observation model when $\tau = 0$. Conversely, when $\tau \to \infty$, the opinions are fully synchronized, thus equivalent to the public assessment model.

Algorithm 5 A pseudo-code for the gossiping model. We use a different ConductGame function from the other models.

```
procedure ConductGame
    for i ← 0 to N − 1 do                                   ▷ Interaction phase
        for j ← 0 to N − 1 do
            if i = j then
                continue
            coopProb ← P_i(m[i][j])
            r ← Random(0, 1)
            if r < coopProb then
                actions[i][j] ← C
            else
                actions[i][j] ← D
    for i ← 0 to N − 1 do                                   ▷ Assessment phase
        for j ← 0 to N − 1 do
            if i = j then
                continue
            k ← Randomly select a player other than i
            goodProb ← R_i(m[i][k], actions[j][k])
            r ← Random(0, 1)
            if r < goodProb then
                assessments[i][j] ← G
            else
                assessments[i][j] ← B
    for i ← 0 to N − 1 do                                   ▷ New assessments are assigned to the image matrix
        for j ← 0 to N − 1 do
            if i = j then
                continue
            m[j][i] ← assessments[j][i]
    for n ← 0 to n_gossip − 1 do                            ▷ Gossiping phase
        i ← Randomly select a player
        j ← Randomly select a player other than i
        k ← Randomly select a player other than i, j
        m[i][k] ← m[j][k]
```

2. Analysis on the solitary observation model

A. Conditional cooperation does not form ESS. In the following, we are going to demonstrate that the conditional cooperator does not generally form ESS in the solitary observation model. This is a more detailed version of the argument in the main text. An intuition is the following: We consider the best action for the donor (Alice, $\mathcal{A}$) against the recipient (Bob, $\mathcal{B}$) under the observation by Charlie ($\mathcal{C}$). In the case of the solitary observation, the opinions toward Bob from Alice and from Charlie are statistically independent. In order to maximize the payoff, Alice does not have to change her actions toward Bob according to her opinion because only Charlie's opinion matters for Alice's long-term payoff. If the long-term benefit of cooperation exceeds the immediate cost of cooperation, Alice should always cooperate irrespective of her opinion. Conversely, Alice should defect unconditionally when the cost of cooperation is higher. Namely, Alice does the free-riding or the second-order free-riding to maximize her payoff.

To support this intuition, let us consider the best response more formally. Assume a sufficiently large population, in which each action is observed by a single observer only. In this case, the probability that j is assessed as G is independent of the observers, m_{ij} is statistically independent of m_{kj} where $i \neq k$. Namely, the image $m_{ij} = 1$ with probability h otherwise 0 for any i, where h is the expected fraction of G image in the image matrix.

First, we consider a mono-morphic resident population in which everyone uses the same norm $\{P, R\}$. Let us denote the average fraction of G in the image as h. Because of the statistical independence, any element of the image matrix m_{ij} is G with probability h and B otherwise. The cooperation probability of the population is written as

$$p_c = hP(G) + (1-h)P(B). \tag{3}$$

The probability that Charlie assesses Alice as G after his observation is

$$h\left[p_c R\left(G,C\right) + \left(1-p_c\right) R\left(G,D\right)\right] + \left(1-h\right)\left[p_c R\left(B,C\right) + \left(1-p_c\right) R\left(B,D\right)\right]. \tag{4}$$

Therefore, the dynamics of h is described as follows:

$$\dot{h} = h^2 R_P\left(G,G\right) + h(1-h)\left[R_P\left(G,B\right) + R_P\left(B,G\right)\right] + \left(1-h\right)^2 R_P\left(B,B\right) - h, \tag{5}$$

where we define

$$R_P(X,Y) \equiv P(X)R(Y,C) + (1-P(X))\,R(Y,D). \quad [6]$$

This is the probability that the donor Alice is assessed as G by the observer Charlie given their original opinions toward the recipient Bob are $m_{\mathcal{AB}} = X$ and $m_{\mathcal{CB}} = Y$, respectively. The first term of Eq. (5) corresponds to the case when both Alice and Charlie assess Bob as G before the interaction. The other terms correspond to the cases for $(m_{\mathcal{AB}}, m_{\mathcal{CB}}) = (G,B)$, (B,G), and (B,B).

When the assessment error rate μ_a is not zero, h reaches a unique stationary solution h^*. The unique stationary solution h^* is obtained by $\dot{h} = 0$:

$$h^* = \begin{cases} \frac{-c_1 - \sqrt{c_1^2 - 4c_2c_0}}{2c_2} & (c_2 \neq 0) \\ -\frac{c_0}{c_1} & (c_2 = 0) \end{cases}, \quad [7]$$

where

$$\begin{aligned} c_2 &\equiv R_P(G,G) - R_P(G,B) - R_P(B,G) + R_P(B,B) \\ c_1 &\equiv R_P(G,B) + R_P(B,G) - 2R_P(B,B) - 1 \\ c_0 &\equiv R_P(B,B) \end{aligned} \quad . \quad [8]$$

Therefore, when a second-order social norm is given, the average reputation h^* and the self-cooperation level are obtained as above. The condition for $h^* = 1$ is obtained as

$$h^* = 1 \iff \begin{cases} R_P(G,G) = 1 \\ R_P(G,B) + R_P(B,G) > 1 \end{cases} \quad [9]$$

In the following, we consider only the stationary state so we denote h^* as h for simplicity.

Next, let us consider the best response. Assume that a small amount of a mutant species having a different norm comes into the system. Suppose that the average cooperation rate of the mutant toward the residents is $p_{\mathrm{mut}\to\mathrm{res}}$. Here, we do not have to define the specific form of the action and the assessment rules for the mutant species because only the mutant's actions matter for the residents to assess the mutant. The mutant may have a complex norm but the mutant is inaccessible to the resident's opinion anyway. Similarly, the residents do not know what the mutant considers. They exchange information only by their actions.

The expected image about the mutant player Alice from the residents H follows the dynamics as

$$\dot{H} = p_{\mathrm{mut}\to\mathrm{res}}\left[hR(G,C) + (1-h)\,R(B,C)\right] + (1-p_{\mathrm{mut}\to\mathrm{res}})\left[hR(G,D) + (1-h)\,R(B,D)\right] - H. \quad [10]$$

After a sufficiently long time, H converges to its stationary value

$$H^* = p_{\mathrm{mut}\to\mathrm{res}}\left[hR(G,C) + (1-h)\,R(B,C)\right] + (1-p_{\mathrm{mut}\to\mathrm{res}})\left[hR(G,D) + (1-h)\,R(B,D)\right]. \quad [11]$$

Below, we denote H^* as H for simplicity. The cooperation probability from the resident to the mutant is

$$\begin{aligned} p_{\mathrm{res}\to\mathrm{mut}} &= HP(G) + (1-H)P(B) \\ &= \left\{p_{\mathrm{mut}\to\mathrm{res}}\left[hR(G,C) + (1-h)\,R(B,C)\right] + (1-p_{\mathrm{mut}\to\mathrm{res}})\left[hR(G,D) + (1-h)\,R(B,D)\right]\right\}P(G) \\ &\quad + \left\{p_{\mathrm{mut}\to\mathrm{res}}\left[h\bar{R}(G,C) + (1-h)\,\bar{R}(B,C)\right] + (1-p_{\mathrm{mut}\to\mathrm{res}})\left[h\bar{R}(G,D) + (1-h)\,\bar{R}(B,D)\right]\right\}P(B) \\ &= p_{\mathrm{mut}\to\mathrm{res}}P_b + P_0, \end{aligned} \quad [12]$$

where $\bar{R}(X,Y) = 1 - R(X,Y)$. Therefore, $p_{\mathrm{res}\to\mathrm{mut}}$ is a linear function of $p_{\mathrm{mut}\to\mathrm{res}}$. Its slope and intercept are

$$P_b \equiv \left[P(G) - P(B)\right]\left\{h\left[R(G,C) - R(G,D)\right] + (1-h)\left[R(B,C) - R(B,D)\right]\right\}, \quad [13]$$

and

$$P_0 \equiv h\left\{R(G,D)\,P(G) + \left[1 - R(G,D)\right]P(B)\right\} + (1-h)\left\{R(B,D)\,P(G) + \left[1 - R(B,D)\right]P(B)\right\}, \quad [14]$$

respectively.

We verified this linear relationship in Eq. (12) with numerical simulations as shown in the main text. This linearity is not limited to the cases when the mutants are the stochastic unconditional cooperators but also holds for more complex classes of strategies, such as the second-order norms. The payoff of the mutant π_{mut} is written as

$$\begin{aligned} \pi_{\mathrm{mut}} &= bp_{\mathrm{res}\to\mathrm{mut}} - cp_{\mathrm{mut}\to\mathrm{res}} \\ &= \left[bP_b - c\right]p_{\mathrm{mut}\to\mathrm{res}} + (\text{terms independent of } p_{\mathrm{mut}\to\mathrm{res}}) \end{aligned} \quad [15]$$

The mutant maximizes its payoff when

$$p_{\mathrm{mut}\to\mathrm{res}} = \begin{cases} 1 & \text{when} \quad bP_b > c \\ \text{any} & \text{when} \quad bP_b = c \\ 0 & \text{when} \quad bP_b < c \end{cases}. \quad [16]$$

Thus, a mutant can maximize its payoff by unconditionally cooperating or defecting. Conditional cooperation does not pay off in general. In other words, the resident species cannot exclude the free riders and the second-order free riders, simultaneously.

Here, bP_b is interpreted as the expected net benefit of cooperation. The first factor of Eq. (13), $[P(G) - P(B)]$, indicates the incentives of being G in the residents' viewpoint. The larger this factor, the more likely the good mutant would be cooperated compared to the case the mutant were B. The second factor $\{h[R(G,C) - R(G,D)] + (1-h)[R(B,C) - R(B,D)]\}$ indicates how more likely the mutant gets G reputation by cooperating.

While the discriminator does not form a Nash equilibrium in general, $bP_b = c$ is the only exception. In this case, the mutant can obtain the same payoff as the residents, thus forming a Nash equilibrium. Generous Scoring (GSCO) norm proposed by Schmid et al. (8) corresponds to this case. GSCO prescribes $P(G) = 1, P(B) = 0, R(-,C) = 1, R(-,D) = 1 - c/(1-2\mu_a)b$ when $N \gg 1$. The rescaled assessment rule $\tilde{R}(-,C) = 1 - \mu_a$ and $\tilde{R}(-,D) = 1 - \mu_a - c/b$ satisfies $bP_b = c$.

The above argument shows that it is not easy to stabilize cooperation in the solitary observation model. This essentially comes from the fact that the mutant has no information about what the observer considers about the recipient since opinions are statistically independent. Therefore, it is essential to introduce the correlation between opinions to overcome the second-order free rider problem.

Norm	$R(G,C)$	$R(G,D)$	$R(N,C)$	$R(N,D)$	$R(B,C)$	$R(B,D)$	$P(G)$	$P(N)$	$P(B)$
GBGGBN-100	G	B	G	G	B	N	1	0	0
GBGGNG-100	G	B	G	G	N	G	1	0	0

Table S1. The definition of the norms used in the ternary reputation model. The assessment rule $R(X,A)$ is the reputation assigned to the donor when the observer's opinion about the recipient is originally X and the donor's action is A. These norms are cooperative ESSs in the public assessment model.

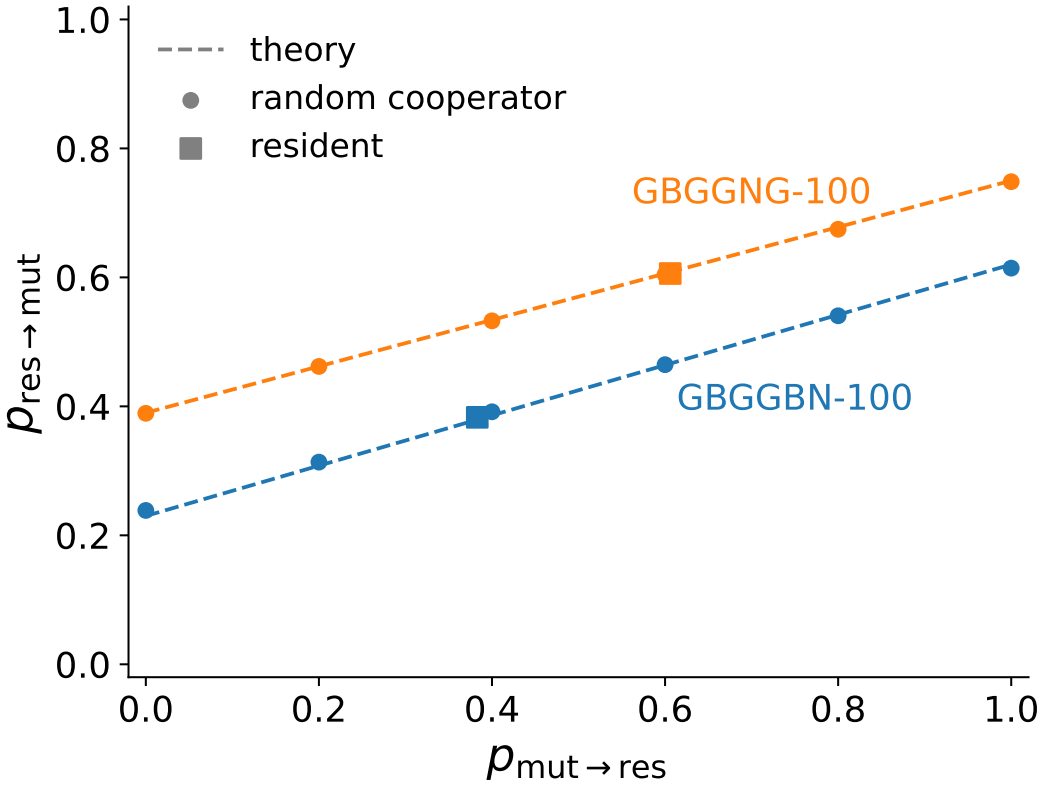


Fig. S1. The cooperation probability of the residents toward the mutants $p_{\mathrm{res}\to\mathrm{mut}}$ as a function of the cooperation probability of the mutants toward the residents $p_{\mathrm{mut}\to\mathrm{res}}$ for the solitary observation model. The circles represent the simulation results when a single mutant, who is a stochastic unconditional cooperator, is introduced. The square represents the cooperation probability of the residents for a population without mutants. The total population size is $N = 100$, and the assessment error rate is $\mu_a = 0.02$. The definition of the resident strategy is shown in Table S1. Theoretical predictions by Eq. (22) are shown as the dashed lines.

B. Non-binary reputation model. While it is a common practice to assume binary reputations, it is straightforward to generalize the conclusion that the solitary observation model cannot sustain evolutionarily stable cooperation. Since this negative conclusion comes from the statistical independence of the opinions, it is not limited to the binary reputation model.

Consider a generalized model where reputations are represented by M different states, $\{S_1, S_2, \ldots, S_M\}$. We define an action rule $\mathbf{P}$ as a column vector that assigns cooperation probabilities for each reputation state, $\mathbf{P} = (P_1, P_2, \ldots, P_M)^{\mathsf{T}}$. For

instance, a donor cooperates with probability P_i when its opinion about the recipient is S_i. Similarly, an assessment rule is represented by a pair of $M \times M$ matrices, $\left\{R^C, R^D\right\}$. The (i,j)-th element of R_C, R^C_{ij}, is defined as the probability that the observer assesses the donor as S_j after observing the donor's cooperation toward the recipient whose reputation was originally S_i from the observer's viewpoint. By definition, the row sum $\sum_j R^C_{ij} = 1$ for all i. In the presence of assessment errors, all elements are neither zero nor one, $0 < R^C_{ij} < 1$. The other matrix R^D is defined similarly for defection.

The reputational state of a monomorphic population is represented by a column vector $\mathbf{h} = (h_1, h_2, \ldots, h_M)^\intercal$, where h_i is the fraction of S_i in the off-diagonal elements of the image matrix. Let us first consider the dynamics of its first element h_1. Its time evolution is

$$\begin{aligned} \dot{h_1} &= \sum_{i=1}^{M}\sum_{j=1}^{M} h_i h_j \left\{P_i R^C_{j1} + [1-P_i]\, R^D_{j1}\right\} - h_1 \\ &= (\mathbf{h}\cdot\mathbf{P})\sum_{j=1}^{M} h_j R^C_{j1} + \mathbf{h}\cdot(\mathbf{1}-\mathbf{P})\sum_{j=1}^{M} h_j R^D_{j1} - h_1, \end{aligned} \tag{17}$$

where $\mathbf{1}$ is the column vector of ones. More generally,

$$\dot{\mathbf{h}} = \left\{(\mathbf{h}\cdot\mathbf{P})\, R^C + [\mathbf{h}\cdot(\mathbf{1}-\mathbf{P})]\, R^D\right\}^\intercal \mathbf{h} - \mathbf{h}. \tag{18}$$

We assume that the population reaches a stationary state $\mathbf{h}^*$ after a sufficiently long time. The stationary state satisfies

$$\mathbf{h}^* = \left\{(\mathbf{h}^*\cdot\mathbf{P})\, R^C + [\mathbf{h}^*\cdot(\mathbf{1}-\mathbf{P})]\, R^D\right\}^\intercal \mathbf{h}^* \tag{19}$$

Hereafter, we consider only the stationary state. We denote $\mathbf{h}^*$ as $\mathbf{h}$ for short. The self-cooperation level is written as the inner product:

$$p_{\mathrm{res}\to\mathrm{res}} = \mathbf{h}\cdot\mathbf{P}. \tag{20}$$

Next, we consider an infinitesimal amount of mutant players with a different norm. Although a mutant may have a different (potentially arbitrarily complex) assessment and action rules, we only need to consider the expected cooperation probability to the residents. This is because the opinion of the mutant is statistically independent of the opinions of the residents. From the resident's viewpoint, only the mutant's cooperation probability matters. Let us say P'_i is the mutant's cooperation probability toward a recipient whose reputation is S_i from a resident observer's viewpoint. Because of the assumption in the solitary observation model, $P'_1 = P'_2 = \cdots = P'_M$ since the opinions of the residents and the mutant are independent. Let us denote this mutant's cooperation probability as $p_{\mathrm{mut}\to\mathrm{res}}$. The reputation of the mutant $\mathbf{H}$ is

$$\mathbf{H} = \left[p_{\mathrm{mut}\to\mathrm{res}} R^C + (1-p_{\mathrm{mut}\to\mathrm{res}})\, R^D\right]^\intercal \mathbf{h}. \tag{21}$$

The cooperation probability from the resident to the mutant is

$$\begin{aligned} p_{\mathrm{res}\to\mathrm{mut}} &= \mathbf{H}\cdot\mathbf{P} \\ &= p_{\mathrm{mut}\to\mathrm{res}}\left[\mathbf{h}^\intercal\left(R^C - R^D\right)\mathbf{P}\right] + \mathbf{h}^\intercal R^D \mathbf{P} \end{aligned} \tag{22}$$

The payoff of the mutant is

$$\begin{aligned} \pi_{\mathrm{mut}} &= b p_{\mathrm{res}\to\mathrm{mut}} - c p_{\mathrm{mut}\to\mathrm{res}} \\ &= p_{\mathrm{mut}\to\mathrm{res}}\left[b\mathbf{h}^\intercal\left(R^C - R^D\right)\mathbf{P} - c\right] + b\mathbf{h}^\intercal R^D\mathbf{P}. \end{aligned} \tag{23}$$

Therefore, π_{mut} is a linear function of $p_{\mathrm{mut}\to\mathrm{res}}$ thus the best action rule for the mutant is

$$\begin{cases} p_{\mathrm{mut}\to\mathrm{res}} = 1 & \text{if } b\mathbf{h}^\intercal\left(R^C - R^D\right)\mathbf{P} - c > 0 \\ p_{\mathrm{mut}\to\mathrm{res}} = \text{any} & \text{if } b\mathbf{h}^\intercal\left(R^C - R^D\right)\mathbf{P} - c = 0 \\ p_{\mathrm{mut}\to\mathrm{res}} = 0 & \text{if } b\mathbf{h}^\intercal\left(R^C - R^D\right)\mathbf{P} - c < 0. \end{cases} \tag{24}$$

Either ALLC or ALLD is the best action rule in general and the conditional cooperation is not evolutionarily stable.

Similar to the case of the binary reputation, this is a consequence of the statistical independence of the opinions. The donor does not have any incentive to cooperate conditionally on its own opinion about the recipient. Because of the statistical independence, the donor's action looks like a random action from the observer's viewpoint.

The above conclusion is valid for any number of reputation states M. The reputation could be represented by ternary values (9), integers (10), or even continuous values (11). Conditional cooperation cannot be evolutionarily stable if the opinions are statistically independent.

To verify the above argument, we conducted numerical simulations for the ternary reputation model (9). The reputations (opinions) are represented by three states, $\{G, N, B\}$. In the case of public assessment, cooperative ESS norms are comprehensively identified, and we use some second-order norms in Table 5 in Ref. (9). The definition of the norms is shown in Table S1. Figure S1 shows the cooperation probability of the residents toward the mutants $p_{\mathrm{res}\to\mathrm{mut}}$ as a function of $p_{\mathrm{mut}\to\mathrm{res}}$. The figure demonstrates that the relationship is linear, and the mutants can unconditionally cooperate or defect to maximize their payoff.

C. Neutral invasion by a random cooperator under solitary observations. The above argument is about the local stability in the vicinity of the monomorphic population. Even when the fraction of the mutants increases, we here show that any norm cannot prevent neutral drift by a random cooperator under solitary observations. Figure S2A shows the payoffs of L3 residents and ALLC mutants as a function of the number of mutant players. While pure L3 is locally unstable against ALLC, L3's payoff surpasses ALLC's as the fraction of the mutants increases, yielding a stable coexistence. By the nature of ALLC, the payoffs of ALLC mutants eventually become less than the residents' payoffs as the fraction of ALLC increases.

However, the population cannot resist neutral invasion by a random cooperator. When the random mutants with the same cooperation probability as the residents, the residents and mutants always have the same payoff, as shown in Fig. S2B. In other words, L3 resident players cannot distinguish other L3 players from those randomly cooperating with the same probability. This is again because of the statistical independence of the opinions. For an L3 observer, the action of other L3 players looks as if it were random. Thus, the random cooperators can neutrally invade the residents. This is not limited to L3 but is a general feature of the solitary observation model. Thus, it is impossible to have ESS with conditional cooperation under solitary observations.

If we do the same analysis for the simultaneous observation model, the results are different. As shown in Fig. S2C, the L3 residents have less payoff than ALLC mutants when the mutants are rare but they reach stable coexistence, similarly to the solitary observation model Fig. S2A. However, the response to the random cooperators is different as shown in Fig. S2D. The L3 residents have higher payoffs than the random cooperators, and the random cooperators cannot invade the residents. In other words, the residents can distinguish the random cooperators from those following the social norm.

To conclude, the solitary observation model cannot sustain evolutionarily stable cooperation. This is because of the independence of the opinions. The conclusion remains valid for arbitrarily complex models, including non-binary reputation models, higher-order norms, and 'dual-reputation update' models (3).

The above argument is based on the assumption that there are at most two types of norms in the population, i.e., the mutation rate is sufficiently small. We do not rule out the possibility that the cooperation can be stable with a mixture of multiple norms. In previous studies for the solitary observation models, it was reported that coexistence between L3 and ALLC is stable against ALLD using the replicator equation (1, 12). Studies on the coexistence of multiple norms are still limited, and it is open whether the coexistence is stable against a wider range of norms.

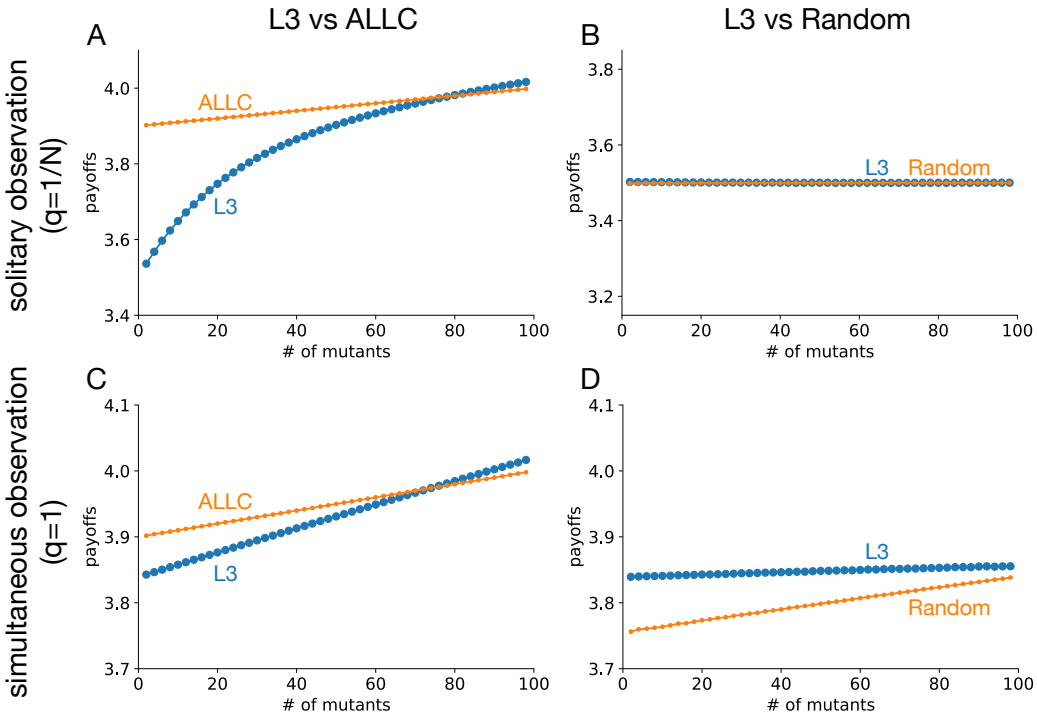


Fig. S2. The payoff comparison between L3 residents vs (A,C) ALLC mutants and (B,D) random mutants. The upper panels are the simulation results for the solitary observation model while the lower panels are simulation results for the simultaneous observation model with $q = 1$. The total population size is $N = 100$ and the assessment error rate is $\mu_a = 0.02$. The cooperation probabilities for the random mutants in (B,D) are the same as the average cooperation probability of the pure L3 population, which are 0.875 and 0.96, respectively.

3. Analysis on the models with correlated opinions

A. Model formulation for the general case. In the following, we discuss the optimal action for Alice when she has a similar opinion about Bob with Charlie. Charlie considers Bob as G with probability h. We consider the conditional probability h_G that Alice also considers Bob as G given Charlie considers Bob as G as well. Similarly, the conditional probability h_B refers to the probability that Alice considers Bob as G given Charlie considers Bob as B.

If Alice's opinion is statistically independent of Charlie's opinion, $h_G = h_B = h$ is satisfied. If Alice's opinion is positively correlated with Charlie's, $h_G > h > h_B$ since Alice is more likely to consider Bob as G when Charlie considers him as G as well. In the perfectly correlated case, namely for the public assessment model, $h_G = 1$ and $h_B = 0$. The anti-correlated case does not happen since these three players are chosen randomly in each round.

Here, we do not introduce any further assumptions on the mechanisms of how their opinions are correlated. Alice and Charlie could have observed Bob's behavior simultaneously, or they could have exchanged their opinions about Bob by gossiping, or a public institution could have disseminated the information about Bob to all players. We just assume that they are statistically correlated while the expected value of the reputation h is not biased by gossiping or whatever.

The probabilities h, h_G, and h_B are not independent of each other. Consider the probability that a pair of randomly chosen players have opposite opinions about a player Bob. These players are written as $h(1-h_G)$ as well as $(1-h)h_B$. Since these must coincide, the following equality must hold:

$$h(1-h_G) = (1-h)h_B \tag{25}$$

Obviously, when $h_G = h$, $h_B = h$ as well. Otherwise, $h_G > h > h_B$.

Next, consider the monomorphic population in which the players use the same norm $\{P, R\}$. The average reputation follows the following dynamics:

$$\dot{h} = hh_G R_P(G,G) + h(1-h_G)R_P(G,B) + (1-h)h_B R_P(B,G) + (1-h)(1-h_B)R_P(B,B) - h \tag{26}$$

Using Eq. (25), the following equation is satisfied in the stationary state:

$$h = hh_G R_P(G,G) + h(1-h_G)\left[R_P(G,B) + R_P(B,G)\right] + (1-2h+hh_G)R_P(B,B) \tag{27}$$

Therefore, given the norm $\{P, R\}$, h is determined once h_G is fixed, and vice versa. There is only one degree of freedom. The value of h (or h_G) depends on the details of the model definitions. Once we define the model to update the reputation (such as solitary observation, simultaneous observation, gossiping, or public assessment) and the norm that the population is using, h as well as h_G and h_B are uniquely determined. For the solitary observation model, where $h = h_G = h_B$, Eq. (27) is identical to Eq. (5).

Depending on the models, the values of h and h_G are not always obtained analytically, so we may need simulations to estimate these values. (For instance, a rigorous analytic solution is yet to be found for the simultaneous observation model with observation probability $q < 1$.) We can calculate h and h_G from simulations by measuring the average and variance of goodness (5). Here, we define the goodness of player-i, g_i, as the fraction of the good image in the i-th column of the image matrix excluding the diagonal element:

$$g_i = \sum_{j\neq i} \delta(m_{ji}, G)/(N-1), \tag{28}$$

where $\delta(x,y)$ is the Kronecker delta function. The average goodness taken over i equals h: $h = \langle g_i \rangle$. The product hh_G is the expected probability that two randomly chosen players agree that another randomly chosen player is G. Thus, for a finite N,

$$\begin{aligned} hh_G &= \left\langle g_i \frac{(N-1)\,g_i - 1}{N-2} \right\rangle = \frac{N-1}{N-2}\left\langle g_i^2 \right\rangle - \frac{\langle g_i \rangle}{N-2} \\ h_G &= \frac{1}{N-2}\left[(N-1)\,\frac{\left\langle g_i^2 \right\rangle}{\langle g_i \rangle} - 1\right]. \end{aligned} \tag{29}$$

When $N \to \infty$, $h_G \to \langle g_i^2 \rangle / \langle g_i \rangle$.

B. Invasion analysis. Next, we consider a mutant having a different action rule P' but the same assessment rule R. We denote Alice's action rule as $P'(X)$ where X is Alice's opinion about Bob. We are going to show that $P(G) = 1$ and $P(B) = 0$ is Alice's best response to maximize her payoff under a certain condition for the resident social norm, $\{P, R\}$, h, and h_G. Namely, Alice's best action rule is the deterministic discriminator under these cases. Otherwise, the best action rule is unconditional cooperator or defector.

We denote the expected opinion from a resident to the mutant as H. The dynamics of H is written as

$$\dot{H} = hh_G R_{P'}(G,D) + h(1-h_G)\left[R_{P'}(G,D) + R_{P'}(B,D)\right] + (1-h)(1-h_B)R_{P'}(B,D) - H, \tag{30}$$

where $R_{P'}(X,Y) = P'(X)R(Y,C) + (1-P'(X))R(Y,D)$. After a sufficiently long time, H converges to its stationary value

$$H^* = hh_G R_{P'}(G,D) + h(1-h_G)\left[R_{P'}(G,D) + R_{P'}(B,D)\right] + (1-2h+hh_G)R_{P'}(B,D), \tag{31}$$

where Eq. (25) is used to simplify the expression. In the following, H^* is denoted as H as we only focus on the stationary state. The cooperation probability from the residents to the mutant is

$$\begin{aligned}
p_{\mathrm{res}\to\mathrm{mut}} &= HP(G) + (1-H)\,P(B) \\
&= \quad hh_G\left\{P'(G)\,R(G,C) + \left[1-P'(G)\right]R(G,D)\right\}P(G) \\
&\quad + hh_G\left\{P'(G)\,[1-R(G,C)] + \left[1-P'(G)\right][1-R(G,D)]\right\}P(B) \\
&\quad + h(1-h_G)\left\{P'(B)\,R(G,C) + \left[1-P'(B)\right]R(G,D)\right\}P(G) \\
&\quad + h(1-h_G)\left\{P'(B)\,[1-R(G,C)] + \left[1-P'(B)\right][1-R(G,D)]\right\}P(B) \\
&\quad + h(1-h_G)\left\{P'(G)\,R(B,C) + \left[1-P'(G)\right]R(B,D)\right\}P(G) \\
&\quad + h(1-h_G)\left\{P'(G)\,[1-R(B,C)] + \left[1-P'(G)\right][1-R(B,D)]\right\}P(B) \\
&\quad + (1-2h+hh_G)\left\{P'(B)\,R(B,C) + \left[1-P'(B)\right]R(B,D)\right\}P(G) \\
&\quad + (1-2h+hh_G)\left\{P'(B)\,[1-R(B,C)] + \left[1-P'(B)\right][1-R(B,D)]\right\}P(B) \\
&= \quad hh_GP'(G)\left\{R(G,C)\,P(G) - R(G,D)\,P(G) - R(G,C)\,P(B) + R(G,D)\,P(B)\right\} \\
&\quad + h(1-h_G)\,P'(B)\left\{R(G,C)\,P(G) - R(G,D)\,P(G) - R(G,C)\,P(B) + R(G,D)\,P(B)\right\} \\
&\quad + h(1-h_G)\,P'(G)\left\{R(B,C)\,P(G) - R(B,D)\,P(G) - R(B,C)\,P(B) + R(B,D)\,P(B)\right\} \\
&\quad + (1-2h+hh_G)\,P'(B)\left\{R(B,C)\,P(G) - R(B,D)\,P(G) - R(B,C)\,P(B) + R(B,D)\,P(B)\right\} \\
&\quad + hh_G\left\{R(G,D)\,P(G) + [1-R(G,D)]\,P(B)\right\} \\
&\quad + h(1-h_G)\left\{R(G,D)\,P(G) + [1-R(G,D)]\,P(B)\right\} \\
&\quad + h(1-h_G)\left\{R(B,D)\,P(G) + [1-R(B,D)]\,P(B)\right\} \\
&\quad + (1-2h+hh_G)\left\{R(B,D)\,P(G) + [1-R(B,D)]\,P(B)\right\} \\
&= \quad hh_GP'(G)\left\{[R(G,C)-R(G,D)]\,[P(G)-P(B)]\right\} \\
&\quad + h(1-h_G)\,P'(B)\left\{[R(G,C)-R(G,D)]\,[P(G)-P(B)]\right\} \\
&\quad + h(1-h_G)\,P'(G)\left\{[R(B,C)-R(B,D)]\,[P(G)-P(B)]\right\} \\
&\quad + (1-2h+hh_G)\,P'(B)\left\{[R(B,C)-R(B,D)]\,[P(G)-P(B)]\right\} \\
&\quad + h\left\{R(G,D)\,P(G) + [1-R(G,D)]\,P(B)\right\} \\
&\quad + (1-h)\left\{R(B,D)\,P(G) + [1-R(B,D)]\,P(B)\right\} \\
&= \quad \left[hh_G\Delta_G + h(1-h_G)\,\Delta_B\right]P'(G) \\
&\quad + \left[h(1-h_G)\,\Delta_G + (1-2h+hh_G)\,\Delta_B\right]P'(B) \\
&\quad + h\left\{R(G,D)\,P(G) + [1-R(G,D)]\,P(B)\right\} + (1-h)\left\{R(B,D)\,P(G) + [1-R(B,D)]\,P(B)\right\}
\end{aligned} \tag{32}$$

where we defined

$$\begin{aligned}
\Delta_G &\equiv [R(G,C)-R(G,D)]\,[P(G)-P(B)] \\
\Delta_B &\equiv [R(B,C)-R(B,D)]\,[P(G)-P(B)]\,.
\end{aligned} \tag{33}$$

These are interpreted as the incentives for cooperation when the recipient is assessed as G or B by a resident observer, respectively, and are dependent on the resident's norm only. The cooperation probability from the mutant to the residents is

$$p_{\mathrm{mut}\to\mathrm{res}} = hP'(G) + (1-h)P'(B) \tag{34}$$

Therefore, the mutant's payoff is

$$\begin{aligned}
\pi_{\mathrm{mut}} &= bp_{\mathrm{res}\to\mathrm{mut}} - cp_{\mathrm{mut}\to\mathrm{res}} \\
&= \alpha_G P'(G) + \alpha_B P'(B) + (\text{terms that are independent of } P',\ h_G, \text{ and } h_B),
\end{aligned} \tag{35}$$

where we defined

$$\begin{aligned}
\alpha_G &\equiv hh_G(b\Delta_G - c) + h(1-h_G)(b\Delta_B - c) \\
\alpha_B &\equiv h(1-h_G)(b\Delta_G - c) + (1-2h+hh_G)(b\Delta_B - c).
\end{aligned} \tag{36}$$

Since $\pi_{\rm mut}$ is a linear function of $P'(G)$ and $P'(B)$, the best action rule $\hat{P}'$ for the mutant is summarized as follows:

$$\hat{P}'(G) = \begin{cases} 1 & \text{when } \alpha_G > 0 \\ \text{any} & \text{when } \alpha_G = 0 \\ 0 & \text{when } \alpha_G < 0 \end{cases}$$
$$\hat{P}'(B) = \begin{cases} 1 & \text{when } \alpha_B > 0 \\ \text{any} & \text{when } \alpha_B = 0 \\ 0 & \text{when } \alpha_B < 0 \end{cases}. \quad [37]$$

From this equation, we conclude the best action rule $\hat{P}'$ is a deterministic action rule unless the special cases of $\alpha_G = 0$ or $\alpha_B = 0$. When α_G and α_B are the same sign, ALLC or ALLD is the best action rule. Conditional cooperation is the best action rule when α_G and α_B have the opposite signs. (Because of the symmetry of G and B, we only consider the case when $\alpha_G > 0$ and $\alpha_B < 0$ without loss of generality in the following.)

When the mutant cannot distinguish Charlie's opinion, namely, $h_G = h_B = h$, these two must have the same sign since α_G is a positive constant multiple of α_B as follows:

$$\frac{\alpha_G}{h} = \frac{\alpha_B}{1-h} = h(b\Delta_G - c) + (1-h)(b\Delta_B - c) \quad [38]$$

Therefore, $\hat{P}'$ is either unconditional cooperation or defection in this case, confirming the conclusion in the previous section.

When α_G and α_B have the opposite signs, the deterministic conditional cooperation $P'(G) = 1$ and $P'(B) = 0$ is the best action. To have $\alpha_G > 0$ and $\alpha_B < 0$, it is necessary that

$$b\Delta_G - c > 0 \quad \text{and} \quad b\Delta_B - c < 0. \quad [39]$$

The left-hand sides of these inequalities ($b\Delta_G - c$ and $b\Delta_B - c$) are the expected extra payoffs from cooperation when the observer's opinion about the recipient is G and B, respectively. If the residents are more able to detect defectors, $b\Delta_G - c$ is larger. If the residents justify the punishment more against bad recipients, $b\Delta_B - c$ gets lower (stronger incentives to defect against bad players.)

When these conditions are satisfied, in order to maximize $\pi_{\rm mut}$, α_G needs to be maximized as well. Thus, mutants are able to obtain a larger payoff with larger h_G and smaller h_B. In other words, the mutant Alice should synchronize her opinions with Charlie's as much as possible.

C. Critical correlation levels for Simple Standing and Stern Judging. Here, we consider specific norms, such as Simple Standing (L3) and Stern Judging (L6), to get more insights.

For L3 norm, substituting $R_P(G,G) = R_P(G,B) = R_P(B,B) = 1 - \mu_a$ and $R_P(B,G) = \mu_a$ into Eq. (27) yields

$$(1-2\mu_a)\, h h_G - 2\,(1-\mu_a)\, h + (1-\mu_a) = 0. \quad [40]$$

Thus, h is determined as a function of h_G as

$$h = \frac{1-\mu_a}{2\,(1-\mu_a) - (1-2\mu_a)\, h_G}. \quad [41]$$

This is an increasing function of h_G. For the solitary observation model, where $h_G = h$, this equation is reduced to a quadratic equation for h. The minimal value of h_G is obtained as its solution:

$$h = \frac{1-\mu_a - \sqrt{\mu_a\,(1-\mu_a)}}{1-2\mu_a}. \quad [42]$$

For instance, $h = h_G = 0.875$ when $\mu_a = 0.02$ for the solitary observation model.

One of the stability condition $\alpha_G > 0$ is then

$$\begin{aligned} h h_G\,(b\Delta_G - c) + h\,(1-h_G)\,(b\Delta_B - c) &> 0 \\ h_G\,(1-2\mu_a)\, b &> c \\ \frac{b}{c} &> \frac{1}{(1-2\mu_a)\, h_G} \end{aligned} \quad [43]$$

using $\Delta_G = 1 - 2\mu_a$ and $\Delta_B = 0$. The other stability condition $\alpha_B < 0$ yields

$$\begin{aligned} h\,(1-h_G)\,(b\Delta_G - c) + (1 - 2h + h h_G)\,(b\Delta_B - c) &< 0 \\ (1-2\mu_a)\, h(1-h_G) b &< (1-h)\, c \\ \frac{b}{c} &< \frac{1-\mu_a - (1-2\mu_a)\, h_G}{(1-2\mu_a)\,(1-\mu_a)\,(1-h_G)}. \end{aligned} \quad [44]$$

From Eq. (40), we can learn some properties of the norms in the small error limit even without further details of the model. When $\mu_a \to 0$, Eq. (40) is reduced to

$$hh_G - 2h + 1 = 0. \tag{45}$$

By definition $h_G \geq h$, thus $hh_G - 2h + 1 = 0 \geq (h-1)^2$. Therefore, $h \to 1$ and $h_G \to 1$ for L3 in the errorless limit whether the model assumes solitary observation, simultaneous observation, or gossiping. This shows that L3 is able to maintain a good reputation even for the private assessment model.

For L6 norm, substituting $R_P(G,G) = R_P(B,B) = 1-\mu_a$ and $R_P(G,B) = R_P(B,G) = \mu_a$ into Eq. (27) yields

$$2\left(1-2\mu_a\right)hh_G + \left(-3+4\mu_a\right)h + \left(1-\mu_a\right) = 0. \tag{46}$$

Thus, the relationship between h and h_G is

$$h = \frac{1-\mu_a}{1+2\left(1-2\mu_a\right)\left(1-h_G\right)}. \tag{47}$$

From this equation, h is an increasing function of h_G. For the solitary observation model, Eq. (46) is reduced to a quadratic equation for h:

$$\left(2h-1\right)\left[\left(1-2\mu_a\right)h - \left(1-\mu_a\right)\right] = 0, \tag{48}$$

whose solution is $h = 1/2$.

The stability condition $\alpha_G > 0$ is written using $\Delta_G = \Delta_B = 1-2\mu_a$ as

$$\begin{aligned} hh_G\left(b\Delta_G - c\right) + h\left(1-h_G\right)\left(b\Delta_B - c\right) &> 0 \\ 2\left(1-2\mu_a\right)\left(2h_G-1\right)b &> c \\ \frac{b}{c} &> \frac{1}{\left(2h_G-1\right)\left(1-2\mu_a\right)}. \end{aligned} \tag{49}$$

The other condition $\alpha_B < 0$ is

$$\begin{aligned} h\left(1-h_G\right)\left(b\Delta_G - c\right) + \left(1-2h+hh_G\right)\left(b\Delta_B - c\right) &> 0 \\ b\left(1-2\mu_a\right)\left(-2hh_G+3h-1\right) &< \left(1-h\right)c \end{aligned} \tag{50}$$

From Eq. (46), $hh_G = \left[\left(3-4\mu_a\right)h - \left(1-\mu_a\right)\right]/2\left(1-2\mu_a\right)$. Therefore, the above inequality is equivalent to

$$b\mu_a\left(1-2h\right) < \left(1-\mu_a\right)c. \tag{51}$$

Since $h \geq 1/2$, the above inequality is always satisfied. L6 is always stable against ALLC.

4. Application to specific models

A. Public assessment model. First, we consider the public assessment model where opinions are perfectly synchronized, namely $h_G = 1$ and $h_B = 0$. More specifically, we consider a model in which a randomly chosen observer assesses the donor. The assessment is publicly announced to all players and all the players agree on the assessment about the donor. This model is equivalent to the public assessment model studied in previous literature (2, 3, 13) when every player has the identical assessment rule.

For the public assessment model, the necessary and sufficient condition for second-order norms to have a strict Nash equilibrium with full cooperation is theoretically obtained. According to Eq. (44) in Ref. (3), the condition is written as:

$$\begin{cases} P(G) = 1 \\ P(B) = 0 \\ R(G,C) = 1 \\ R(B,D) > 0 \\ R(G,D) < 1 \\ \frac{b}{c} > \frac{1}{1-R(G,D)} \\ R(B,C) \leq R(B,D) \quad \text{or} \quad \frac{b}{c} < \frac{1}{R(B,C)-R(B,D)} \end{cases}. \tag{52}$$

This is consistent with Eq. (37). Mutant's best action rule is the deterministic discriminator $P(G) = 1$ and $P(B) = 0$ when

$$\begin{cases} \alpha_G &= h(b\Delta_G - c) > 0 \\ \alpha_B &= (1-h)(b\Delta_B - c) < 0 \end{cases}. \tag{53}$$

Since $0 < h < 1$, P is the unique best action rule when

$$\begin{cases} b\left[R\left(G,C\right) - R\left(G,D\right)\right] > c \\ b\left[R\left(B,C\right) - R\left(B,D\right)\right] < c \end{cases}, \tag{54}$$

which agrees with the last three inequalities in Eq. (52). Note that the other conditions in Eq. (52) came from the full-cooperation condition.

B. Simultaneous observation model. Next, we consider a model where multiple observers assess the donor simultaneously. When the donor cooperates or defects against the recipient, the other players observe the donor's action with probability q. In the limit of $q \to 1/N$, the model falls back to the solitary observation model. This model has been studied theoretically and numerically (4–6, 14).

An approximate analytical result is obtained for $q = 1$ (5, 6, 15). They analytically calculated the distribution of the goodness in the stationary state by approximating it as a superposition of Gaussian functions. They concluded that Simple Standing (L3) norm promotes stable cooperation against other second-order norms for a range of b/c. When the error rate $\mu_a \ll 1$ and the population size $N \to \infty$, the distribution of the goodness of the L3 residents is written as the sum of delta functions (See Fig. 3D in Ref. (6)):

$$p(g) = (1-2\mu_a)\delta\left(g-(1-\mu_a)\right) + \mu_a\delta\left(g-(1-3\mu_a)\right) + \mu_a\delta\left(g-2\mu_a\right) + O(\mu_a^2). \quad [55]$$

From their results, L3 is stable against ALLC and ALLD for $1 < b/c < 2$ when the implementation error rate tends to zero (15).

From Eq. (55) and Eq. (29), we calculate h and h_G as

$$\begin{aligned} h &= \int_0^1 gp(g)dg = 1-2\mu_a + O(\mu_a^2), \\ h_G &= \frac{\int_0^1 g^2 p(g)dg}{\int_0^1 gp(g)dg} = 1-\mu_a + O(\mu_a^2). \end{aligned} \quad [56]$$

These are consistent with Eq. (27), validating the calculation. According to Eq. (37), the best action rule for the mutant is the deterministic discriminator when

$$\begin{cases} \alpha_G > 0 & \iff \frac{b}{c} > 1 + O(\mu_a) \\ \alpha_B < 0 & \iff \frac{b}{c} < 2 + O(\mu_a) \end{cases}, \quad [57]$$

reproducing the results by Fujimoto and Ohtsuki (6).

An analogous calculation can be done for other norms. For Stern Judging (L6) norm, the goodness distribution has a single peak at $g = 1/2$ (6):

$$p(g) = \delta(g - 1/2), \quad [58]$$

which yields $h = h_G = 1/2$. Thus, we obtain $\alpha_G < 0$ and $\alpha_B < 0$, irrespective of b and c. Therefore, unconditional defection is the best action rule for the mutant, indicating that L6 cannot stabilize cooperation in this model.

In addition to these two norms, let us consider another set of norms that interpolates between L3 and L6, namely, $R(G,C) = R(B,D) = 1$, $R(G,D) = 0$, while $R(B,C)$ is varied in $[0,1]$. When $R(B,C) = 0$ and $R(B,C) = 1$, the norm is equivalent to L6 and L3, respectively. Figure S3A shows h and h_G as functions of $R(B,C)$. Since no analytical solution exists for h and h_G, we calculated these values using Monte Carlo simulations. As $R(B,C)$ increases, both h and h_G increase quickly. Even with $R(B,C) \approx 0.5$, h goes as high as 0.9, showing a high self-cooperation level. Figure S3B shows the ranges of b/c for which the discriminator action rule is stable. Although no stable cooperation exists for L6, the lower bound of b/c quickly drops as $R(B,C)$ increases, yielding a wide range of b/c for stable cooperation. As $R(B,C)$ approaches 1, the upper bound of b/c also decreases, narrowing the stable range. L6 is too strict to stabilize cooperation, while L3 is too permissive to prevent invasion by ALLC. In the intermediate region, the norm is more tolerant against ALLC while maintaining a reasonably high self-cooperation level. These theoretical results are reproduced by Monte Carlo simulations, confirming the validity of our framework.

As another demonstration, we consider the effect of the observation probability q. Figure S3C shows h and h_G as functions of q for strategies with $R(G,C) = R(B,D) = 1$, $R(G,D) = 0$, and $R(B,C) = 0.5$. As q decreases to zero, h and h_G converge to the same value, reproducing the solitary observation model. The stable range of b/c for the discriminator strategy is shown in Fig. S3D. Again, the stable range narrows as q decreases and eventually vanishes when $q \to 0$, indicating that the solitary observation model cannot stabilize cooperation.

C. Gossiping Model. Another application of the above theory is the gossiping model proposed by Kawakatsu et al. (7). In their model, the gossiping process is introduced in the solitary observation model. After a round of the donation games, observers assess the donors' actions as in the solitary observation model. The gossiping process follows the private assessment. Each player consults a randomly selected peer at a certain interval and adopts her view about the donor. Donation games, private observations, and gossiping are repeated alternately until the reputations equilibrate. The gossip dynamics for a focal individual are therefore described by a bi-allelic Wright-Fisher process, which keeps track of how many individuals view the focal individual as good or bad over discrete generations of gossip.

The number of gossip rounds quantifies the amount of peer-to-peer gossip between private observation periods. The gossip duration τ quantifies the level of agreement between the players. When $\tau \to 0$, the gossiping process is equivalent to the solitary observation model, while $\tau \to \infty$ corresponds to the public assessment model.

Here, we show a more detailed derivation of the critical gossip duration for the gossip model following the previous paper (7) with our notations. As shown below, we obtain the two equalities of h and h_G by considering the reputation dynamics driven by assessments and gossiping. These two equalities determine h and h_G after the equilibration of the reputations. Then, we consider an infinitesimal amount of ALLD mutants and the stability of the discriminator strategy against the mutants by

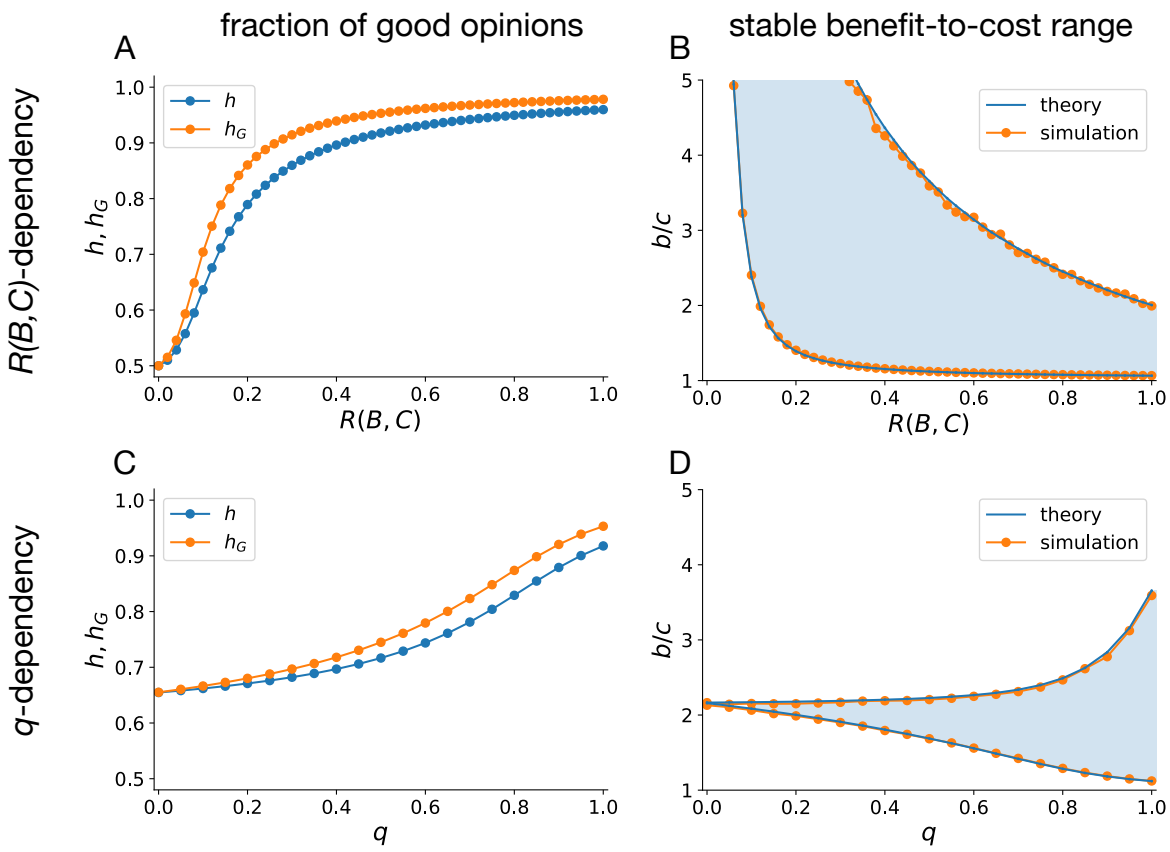


Fig. S3. (A, B) An interpolation between the Simple Standing (L3) and Stern Judging (L6) norms for the simultaneous observation model. A norm is defined by $R(G,C) = R(B,D) = 1, R(G,D) = 0$, and $R(B,C)$ is controlled. (A) The relationships between h and h_G calculated by Monte Carlo simulation for monomorphic populations as functions of $R(B,C)$. (B) The right panel shows the stable range of b/c for the discriminator strategy. The blue lines indicate the critical b/c calculated by the theoretical conditions $\alpha_G > 0$ and $\alpha_B < 0$. To obtain these lines, we used h and h_G shown in (A). The red points indicate the same values calculated by Monte Carlo simulations by introducing a single ALLC/ALLD mutant. We introduced a single ALLC/ALLD mutant individual and calculated the average cooperation probabilities. We estimated the critical b/c as the point where the payoff of the discriminator strategy is equal to that of the mutant. (C, D) The same plots as (A, B) but for different observation probability q. $R(B,C) = 0.5$ is fixed. As $q \to 0$, the results converge to the solitary observation model. The population size $N = 100$ and the error rate $\mu_a = 0.02$. The results are averaged over 100 independent runs.

comparing their payoffs. Hereafter, we only consider L6 (Stern Judging) norm with assessment error rate μ_a, but analogous calculations can be done for other norms as shown in (7).

We consider the reputation dynamics driven by gossiping. By gossiping, the agreement level of the reputation increases while the average reputation remains unchanged. According to Ref. (7), the (dis)agreement levels after gossiping are updated according to the Wright-Fisher process. They denote $\widetilde{g_2}$ ($\widetilde{b_2}$) as the probability that two randomly selected players agree that a third individual is G (B) after gossiping. These variables indicate the extent of agreement of the reputations. They also introduced $2\widetilde{d_2}$, which is the probability that two randomly selected players have opposite opinions about a third individual. By definition, $\widetilde{g_2} + \widetilde{b_2} + 2\widetilde{d_2} = 1$. In our notation, these probabilities are expressed using h and h_G as follows:

$$\begin{aligned} \widetilde{g_2} &= hh_G \\ \widetilde{b_2} &= (1 - 2h + hh_G) \\ \widetilde{d_2} &= h(1 - h_G). \end{aligned} \tag{59}$$

Since these variables follow the Wright-Fisher process starting from the initial states $\widetilde{g_2} = h^2$, $\widetilde{b_2} = (1-h)^2$, and $\widetilde{d_2} = h(1-h)$, the agreement levels are updated as

$$\begin{aligned} \widetilde{g_2} &= h^2 + h(1-h)\left(1 - e^{-\tau}\right), \\ \widetilde{b_2} &= (1-h)^2 + h(1-h)\left(1 - e^{-\tau}\right), \\ \widetilde{d_2} &= h(1-h)e^{-\tau}, \end{aligned} \tag{60}$$

where τ is the scaled gossip duration. From these, we obtain a simple relationship between h and h_G after gossiping as

$$1 - h_G = (1-h)e^{-\tau}. \tag{61}$$

Combining Eq. (61) and Eq. (47), we obtain the relationship between τ and h as

$$\begin{aligned} 2\left(1-2\mu_a\right)h(1-h)e^{-\tau} &= 1-h-\mu_a \\ e^{\tau} &= \frac{2\left(1-2\mu_a\right)h\left(1-h\right)}{1-h-\mu_a} \\ \tau &= \log\left[\frac{2\left(1-2\mu_a\right)h\left(1-h\right)}{1-h-\mu_a}\right]. \end{aligned} \tag{62}$$

From the above, it is straightforward to show that τ is an increasing function of h when $\mu_a \approx 0$.

Lastly, we consider the stability of the discriminator strategy against an infinitesimal amount of ALLD mutants. The average reputation of the ALLD mutant is $\mu_a h + (1-\mu_a)(1-h)$ since it gets G with probability μ_a and $1-\mu_a$ when it meets G and B recipients, respectively. The payoff of an ALLD player is thus $b\left[\mu_a h + (1-\mu_a)(1-h)\right]$. On the other hand, the residents' payoff is $(b-c)h$. The residents are stable if and only if

$$\begin{aligned} b\left[\mu_a h + (1-\mu_a)(1-h)\right] &< (b-c)h, \\ \left[2\left(\frac{b}{c}\right)(1-\mu_a) - 1\right]h &> \frac{b}{c}(1-\mu_a) \end{aligned} \tag{63}$$

. From this condition, Kawakatsu et al. (7) derived the critical gossip duration τ^* as

$$\tau^* = \log\left[\left(2 - \frac{\frac{b}{c}}{\frac{b}{c} - \frac{1}{2(1-\mu_a)}}\right)\left(\frac{\frac{b}{c}}{\frac{b}{c} - \frac{1}{1-2\mu_a}}\right)\right]. \tag{64}$$

The same conclusion on the critical gossip duration is also obtained from our framework. For L6, the condition $\alpha_G > 0$ is calculated as in Eq. (49). Plugging Eq. (47) into Eq. (49), we obtain

$$\begin{aligned} (2h-1)(1-\mu_a)b &> ch \\ \left[2\frac{b}{c}(1-\mu_a) - 1\right]h &> \frac{b}{c}(1-\mu_a), \end{aligned} \tag{65}$$

which is identical to Eq. (63). Thus, our theory is equivalent to the previous study in terms of the stability of the discriminator strategy.

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